

Dune Rider - Final Game Project

For my final game project, Dune Rider, I extended the base game by adding moving scorpion enemies and a sound system. I used a constructor function to create each scorpion with its own starting position, movement range and speed. The scorpions are stored in an array and updated using a loop. They patrol back and forth, change direction when reaching the end of their range, and use collision detection to determine when they touch the player. I also added different sound effects for jumping, collecting water bottles, losing a life, falling into a canyon, winning and reaching game over. Background music was added to make the game feel more complete.

The most difficult parts of the project were getting the movement and game-state logic to work reliably. Jumping, gravity and falling into pits were challenging because several conditions had to work together without conflicting. I also found it difficult to make the side-scrolling camera follow the player while keeping the HUD fixed on the screen. Adding sounds required troubleshooting issues such as audio delay, volume and preventing different sounds from overlapping.

Throughout the project, I practised using arrays, loops, objects, functions, constructor functions and conditional statements. I also improved my understanding of transformations, collision detection, game states and sound handling. The project also helped me improve my debugging skills by testing features individually and fixing problems before moving on.